Grocery Auto Planner

Complete PRD • System Design • Docker-only Implementation Blueprint

Web MVP first, React Native extension later. Startup-style documentation suitable as a Cursor AI implementation guide.

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Executive Summary

Build an AI-powered grocery planning assistant that orchestrates multiple MCP servers through LangGraph.

Implementation Notes

- Docker-only development using Docker Compose.
- FastAPI gateway exposes REST APIs.
- LangGraph orchestrates independent MCP servers.
- PostgreSQL stores durable state.
- Redis accelerates repeated lookups.
- Kafka carries asynchronous events.

Vision

Deliver a web-first AI assistant that plans meals, manages pantry inventory and automates grocery decisions.

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Problem Statement

Users repeatedly buy duplicate groceries, waste food and spend time comparing prices.

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Goals & KPIs

Reduce planning time below 30 seconds, pantry accuracy above 95%, and improve weekly retention.

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User Personas

Working professionals, students, families and home chefs.

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Core Features

Meal planning, OCR receipts, pantry tracking, shopping lists, reminders, analytics and authentication.

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High-Level Architecture

React → FastAPI Gateway → LangGraph Planner → MCP Servers → PostgreSQL/Redis/Kafka.

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Core Services

Service	Purpose
frontend	React Web App
gateway	FastAPI API Gateway
planner	LangGraph Orchestrator
recipe-mcp	Recipe Tools
pantry-mcp	Inventory Tools
ocr-mcp	Receipt OCR
price-mcp	Store Comparison
redis	Caching
postgres	Database
kafka	Events
minio	Receipt Storage

MCP Servers

Recipe, Pantry, OCR, Price Comparison, Reminder, Calendar and Maps servers.

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LangGraph Workflow

Planner coordinates every MCP tool while maintaining workflow state.

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Database Design

Normalized PostgreSQL schema with pantry_items, recipes, shopping_lists, receipts and orders.

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API Design

REST endpoints for auth, pantry, planner, receipts and MCP discovery.

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Docker Architecture

Every component runs inside Docker Compose with no local service installations.

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Security

JWT, RBAC, HTTPS, audit logging and secret management.

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Redis Strategy

Cache inventory, recipes and comparisons.

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Kafka Events

receipt_processed, pantry_updated, shopping_list_generated and order_completed.

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OCR Pipeline

Receipt → MinIO → OCR MCP → Pantry Update.

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Price Comparison Engine

Normalize responses and optimize delivery-aware recommendations.

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Frontend UX

Dashboard, Pantry, Meal Planner, Receipt Scanner and Shopping List.

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Deployment

Nginx reverse proxy with Docker Compose on a VPS.

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Testing

Unit, integration, MCP contract and end-to-end tests.

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Monitoring

Prometheus, Grafana and structured logging.

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Roadmap

MVP → OCR → Family Mode → Autonomous Ordering → Mobile.

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Folder Structure

Monorepo with frontend, gateway, planner and independent MCP services.

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Cursor Kickstart Prompts

Generate Docker-first services, typed APIs and tests.

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Future Vision

Smart fridge integration, nutrition planning and voice-first mobile assistant.

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Appendix

Suggested repository: frontend/ gateway/ planner/ mcp-services/ infra/ docs/ tests/ docker-compose.yml

Demo flow: Login → Scan Receipt → Pantry Updated → Ask for Biryani → Receive Shopping List → Compare Stores → Schedule Reminder.